National Taipei University of Technology

Object-Oriented Analysis and Design

Homework #5

In homework 5 you are required to: (1) revise homework 4 **based on the comments given in the previous homework (if any)** and (2) **implement the use case(s)** you designed in homework 4.

The implementation activities include programming (writing production code) and unit tests (using xUnit framework). That is, in this homework you should provide source code listing and testing code listing (see section 6 and 7, respectively). Snapshots, such as the execution of your system and the result of unit testing, may be provided as well. You also have to construct the implementation class model in section 5, and show the difference between the new implementation class model and the design class model you made in homework 4.

NOTE: *For two-person teams or three-person teams, you should implement more use cases in this homework, or you may not complete the entire system before the end of the course. The actual number of uses cases that you need to implement depends on the scale and difficulty of the use cases you choose. If the use cases are relatively small and easy, you should implement a few more use cases.*

1 Cover Page

The cover page includes the *project title*, *homework number*, and *team members*.

2 [5%] Requirement Document

2.1 Change History

2.2 Problem Statement

2.3 Summary of System Features [Larman04, Ch7.7]

2.4 Use Case Diagram [Larman04, Ch6.17]

2.5 Use Cases [Larman04, Ch6]

2.6 Non-functional Requirements and Constraints [Larman04, Ch7.4]

2.7 Glossary [Larman04, Ch7.9]

2.8 Software Environments

3 [5%] Domain model

3.1 Domain class diagram showing only concepts [Larman04, Ch 9.4 ~ 9.6]

3.2 Add associations [Larman04, Ch 9.14]

3.3 Add attributes [Larman04, Ch 9.16]

4 [15%] Design

4.1 Logical Architecture with UML package diagram [Larman04, Ch 13]

4.2 Use-Case Realizations with GRASP Patterns [Larman04, Ch 17]

4.2.1 Choose the <u>most significant use-case</u>, identify system events, and draw a system sequence diagram for that use case.

4.2.2 Draw a sequence diagram <u>for each</u> system event you identified, and label with the GRASP patterns to describe how you assign responsibilities.

4.3 Design Class Model (domain class model after class design) [Larman04, Ch 16]

5 [15%] Implementation Class Model (new in homework 5)

5.1 Draw an implementation class diagram for your system (including associations, attributes and methods)

5.2 Show the difference between implementation class model and design class model

Construct the following two tables:

Table 5.2.1: Comparison with design and implementation class

Class	Method	Design	Imp.
ClassName1	m1	*Yes*	*Yes*
	m2	*No*	*Yes*
ClassName2 (added in design phase)	m1	*No*	*Yes*
	m2	*No*	*Yes*
ClassName3 (removed in design phase)	m1	*Yes*	*No*
	m2	*Yes*	*No*

Table 5.2.2: Summary of implementation class/method changed

	Number of added	Number of removed	Number of modified
Class			
Method			

Note: Number of modified means the number of modification of signature.

5.3 Calculate Line of Code

Provide the following table for your implementation classes:

Table 5.3.1: Line of Code of classes

No	Class Name	Number of methods	Line of Code in Class (without comment)
1			
2			
…			
Sum			

6 [15%] Programming (new in homework 5)

6.1 Snapshots of system execution

6.2 Source Code Listing (note: select and list the most important source code; no more than 10 pages)

7 [15%] Unit Testing (new in homework 5)

7.1 Snapshots of testing result

7.2 Unit Testing Code Listing (note: select and list the most important test code; no more than 10 pages)

8 [30%] Revise your requirement document, domain model, and design based on TA's comments or your own findings, if any. In case that the grading of HW#4 is not completed yet, you should contact TA and obtain comments before submitting this assignment.

9 Measurement

How many hours did you spend for this homework? Please record the time precisely in "`YY/MM/DD HH:mm ~ HH:mm`" format. **Keep logging precisely for each team member.**